

CONTENTS

- A

Executive Summary
- B

Cyber & Economic Exposure Monitor
- C

Data Refresh & Changelog
- D

Deep-Dive: Monthly Theme
- E

European Context Card
- F

SSI Enhanced Neural Network
- F.5

Anti-maladaptation Gate (Radar 2 · W1–W10) v4.2
- G

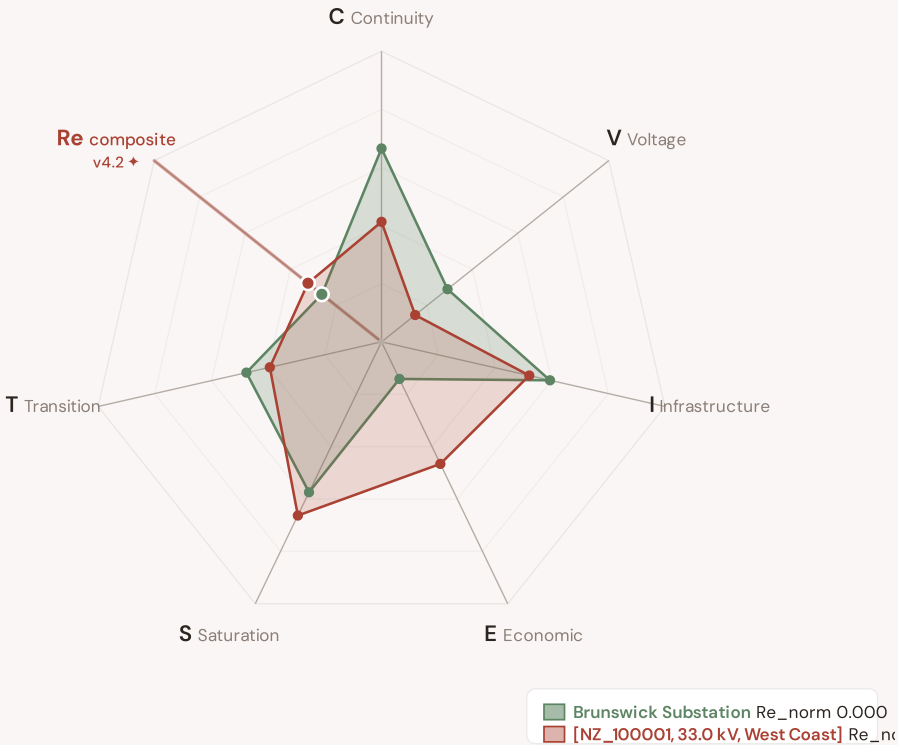
Looking Ahead

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar ($Re_norm \in [0, 1]$ via $clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$). Two worked examples overlaid: **Brunswick Substation** (low-exposure reference, Re_norm 0.000) · **[NZ_100001, 33.0 kV, West Coast]** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds $[0.920, 1.787]$. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

1,589

91 EHV · 110 HV · 1,388
distribution-tier

76,379

~14,551 km coverage

42.9 M

10,000 per substation

34

101 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 1,589 substations and 76,379 grid lines spanning ~14,551 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering New Zealand's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Brookside zone substation

Canterbury, Canterbury · 66 kV

0.871

Critical

0.693

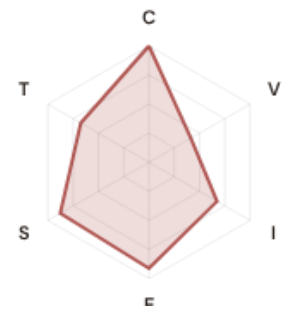
89th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



Re composite v4.2 7th axis

Re_raw 1.1598 Re_norm 0.359

MODERATE EXPOSURE



COMPONENTS

C Continuity: 1.000 / 0.30 (333%) ▲

I Infrastructure: 0.673 / 0.25 (269%)

S Saturation: 0.877 / 0.20 (438%)

V Voltage: 0.413 / 0.10 (413%)

E Economic: 0.912 / 0.10 (913%)

T Transition: 0.674 / 0.05 (1347%)

MODIFIERS

R3 Consequence: 0.984 ↓ dampens

R4 Graph Criticality: 0.990 → neutral

R6a Restoration: 1.000 → neutral

R6b Seismic: 1.192 ↑ amplifies

R7 Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for September 2026 — is **Brookside zone substation** in Canterbury, Canterbury. With an R_final of 0.871 (Critical band, 89th fleet percentile), its risk profile is dominated by the T (Transition) component at 1347% of its weight ceiling, followed by E (Economic) at 913%. Modifiers collectively shift R_base by 6.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

R7 v2 CRA-anchored transition (Q3 2026 → Q1 2027 dual-write): the R7 signal is entering a dual-write window where the legacy R7_cyber (DESI + ACN scalar proxy) co-exists with R7_cyber_v2 (CRA Article 14 + NIS2 Article 21 register-anchored composite, Path C+D). Consumers may select either; substation audit trail carries _r7_cyber_v2_source when v2 populated.

CYBER HIGH EXPOSURE

O

0.0% of fleet

BLIND SPOTS

427

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

0.9965

Range [0.99, 1.05]

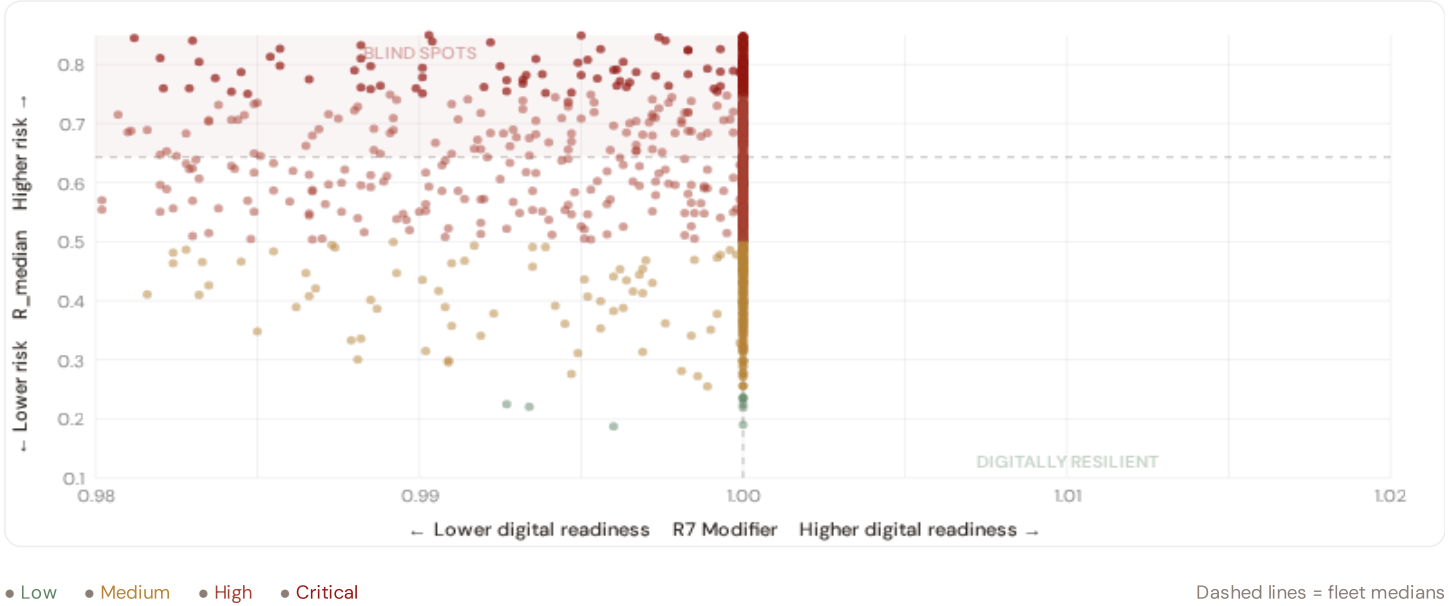
HIGH-CONSEQUENCE SUBS

221

13.9% · R3 ≥ 1.03

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



The cyber-physical matrix identifies **427 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Wellington (117), Canterbury (103), Auckland (71)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Substations serving highest-value economic activity — pharma / chemicals / data centres / financial districts / heavy industry

156 substations (9.8%)

High/Crit/Ext: 145 (92.9%)

Avg R: 0.718

Avg E: 0.972 / 0.10

Low 0

Med 11

High 76

Crit 64

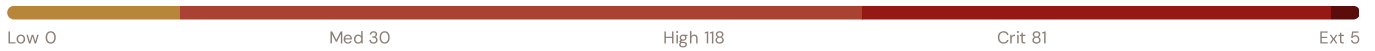
Ext 5

Industrial / Medium–Large Enterprise

Est. VoLL: €8–15/kWh

Industrial belt — significant continuity requirements

234 substations (14.7%) High/Crit/Ext: **204** (87.2%) Avg R: **0.691** Avg E: **0.778** / 0.10



Commercial / SME–Dense

Est. VoLL: €3–8/kWh

Service economy, retail, urban commercial districts — moderate individual impact

389 substations (24.5%) High/Crit/Ext: **335** (86.1%) Avg R: **0.677** Avg E: **0.584** / 0.10

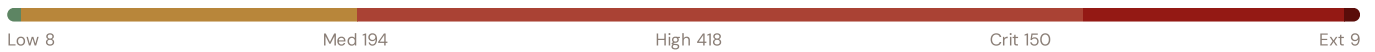


Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse activity, agricultural land — lower VoLL

779 substations (49.0%) High/Crit/Ext: **577** (74.1%) Avg R: **0.614** Avg E: **0.264** / 0.10



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR-E) segments substations by the economic value they serve. **145 substations** combine capital-intensive economic fabric (top-decile Economic component ($E \geq 0.891$, P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Wellington (42), Auckland (30), Canterbury (24)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 6.1%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.

R7 Digital Readiness
Low High

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Gisborne ▲ high R	0.9944
2	West Coast ▲ high R	0.9947
3	Taranaki	0.9957
4	Wellington ▲ high R	0.9958
5	Bay of Plenty ▲ high R	0.9960
6	Canterbury ▲ high R	0.9960
7	Marlborough	0.9962
8	Auckland	0.9965
9	Hawke's Bay ▲ high R	0.9967
10	Manawatu–Whanganui	0.9973
11	Waikato	0.9973
12	Otago	0.9978
13	Southland	0.9979

14	Tasman		0.9981
15	Northland		0.9985

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Gisborne** has the lowest average R7 (0.9944), while **Northland** leads at 0.9985 — a spread of 0.0042 across the R7 range. **6 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (September 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9965, median = 1.0000; 0 substations classified HIGH cyber-exposure; 427 blind-spot substations (High/Critical/Extreme risk + below-median R7); 156 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI’s credibility.

TOTAL SOURCES

— variables

HOURLY — MONTHLY

O

High-frequency feeds

QUARTERLY / ANNUAL

O

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

Data Source Registry — September 2026

OSM	OpenStreetMap — Power Infrastructure	WEEKLY	Node/edge	3 vars
EA	Electricity Authority (EA)	MONTHLY	National	5 vars
EECA	EECA (Energy Efficiency & Conservation Authority)	QUARTERLY	Regional	4 vars
NZTA	NZTA (NZ Transport Agency)	QUARTERLY	Regional	1 var
MBIE	MBIE (Ministry of Business, Innovation & Employment)	ANNUAL	Regional	4 vars
ME	Ministry for the Environment	ANNUAL	Regional	3 vars
CC	Commerce Commission	ANNUAL	TA/Regional	3 vars
StatsNZ	Stats NZ (Tatauranga Aotearoa)	ANNUAL	TA	8 vars
Vector	Vector Ltd — Largest EDB	ANNUAL	Network	2 vars
Orion	Orion NZ — Canterbury/Christchurch Distribution	ANNUAL	Network	2 vars
Powerco	Powerco — Waikato/Taranaki/Manawatu Distribution	ANNUAL	Network	2 vars
Meridian	Meridian Energy — Largest Generator	ANNUAL	Regional	2 vars
Genesis	Genesis Energy — Thermal/Hydro/Wind	ANNUAL	Regional	2 vars
Contact	Contact Energy — Geothermal/Gas Generation	ANNUAL	Regional	2 vars
Mercury	Mercury NZ — Waikato Hydro/Geothermal	ANNUAL	Regional	2 vars
FE-NZ	Fire and Emergency NZ	ANNUAL	Regional	2 vars
EQC	Earthquake Commission (EQC)	ANNUAL	Regional	2 vars
LGNZ	LGNZ (Local Government NZ)	ANNUAL	TA	2 vars
BRANZ	BRANZ (Building Research Association)	ANNUAL	National	2 vars
EDB-Disc	EDB Information Disclosures	ANNUAL	EDB	3 vars
LINZ	LINZ (Land Information New Zealand)	STATIC	National	3 vars
Copernicus	ERA5 / Copernicus Climate Data Store	STATIC	0.25° (~27 km)	4 vars
GeoTech	NZ Geotechnical Database	STATIC	Site-level	2 vars
TPNZ	Transpower NZ — TSO	REAL-TIME	GXP/Regional	8 vars
NIWA	NIWA (National Institute of Water & Atmospheric Research)	HOURLY	~25 km	4 vars
GeoNet	GNS Science / GeoNet	REAL-TIME	Event-level	3 vars
SO	Transpower System Operator (SO)	REAL-TIME	Real-time	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE

1,589

substations scored

MEDIAN R

0.643

P5=0.332 · P95=0.933

HIGH + CRITICAL + EXTREME

1261

79.4% of fleet

HIGH CONFIDENCE

95%

1505 subs · CI/R < 0.50



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,589 substation scores remain unchanged** this edition. The fleet median R of 0.643 (mean 0.639) reflects a distribution skewed toward the lower bands, with 0.6% in Low and 18.1% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, PI, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	\$5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	\$5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	\$5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	\$5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	\$5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	\$5
V42-7	NEW	Re composite — bounded resilience scalar Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00); Re_norm = clip((Re_raw - 0.920)/(1.787 - 0.920), 0, 1)	\$7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	\$14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	\$4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data

SECTION D — MONTHLY DEEP-DIVE

The Auckland North Island Backbone

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT
SUBSTATIONS

342

9 EHV · 333 HV

HIGH BAND

120 + 179 +
30

96.2% of corridor

CORRIDOR MEDIAN R

0.790

Fleet median: 0.643

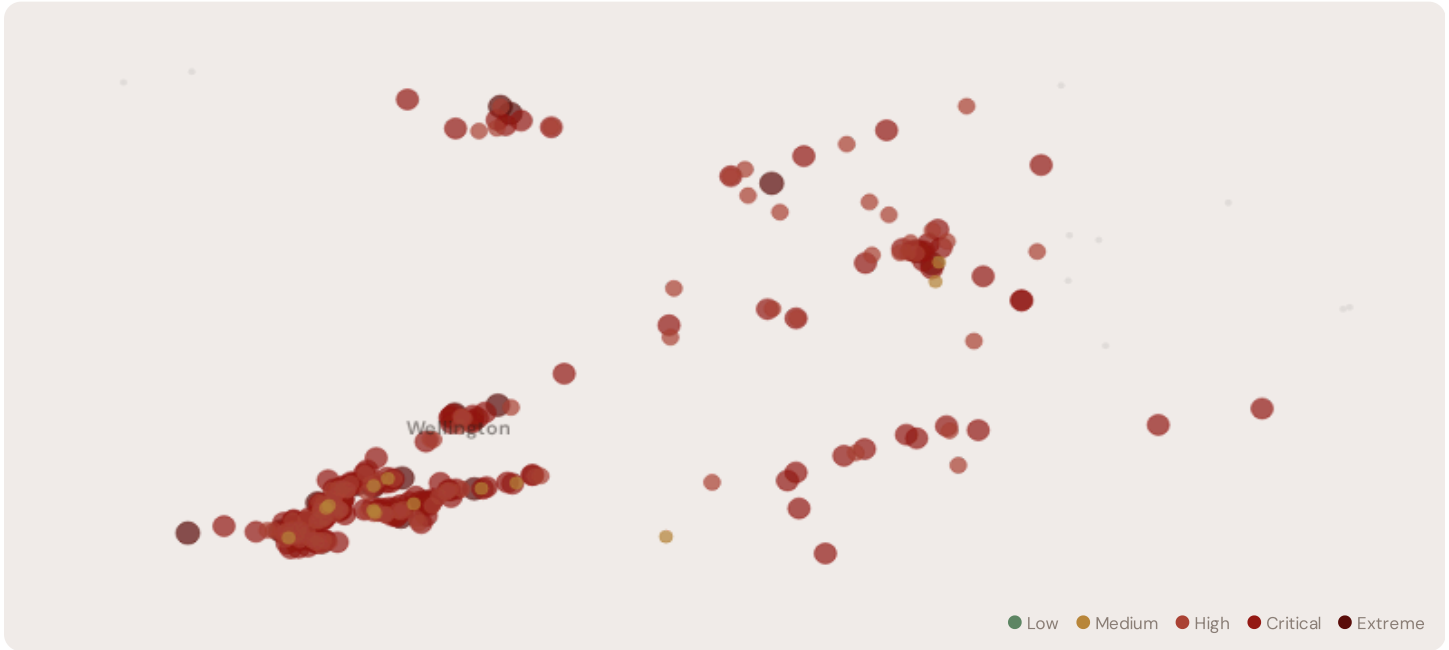
WORST PROVINCE

Wellington

Avg R: 0.790 · 342
substations

The Wellington hosts **342 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **120 substations (35.1%)** are classified High and **179** Critical, with only **0** in the Low band. 81% of corridor substations score above the national fleet median of 0.643. The corridor median R of 0.790 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Chaytor Street Substation	Wellington	1.191	Extreme	1.000	0.825	0.739	0.221	0.000	0.541	C	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Newtown Zone Substation (Decommissioned)	Wellington	1.172	Extreme	0.439	0.708	0.854	0.520	0.430	0.347		
Manawatu– Whanganui_516956484	Wellington	1.172	Extreme	0.774	0.251	1.000	0.646	0.479	0.436	I	
Manawatu– Whanganui_1097458905	Wellington	1.151	Extreme	0.930	0.814	0.573	1.000	0.192	0.380	C	
Marlborough_1093037035	Wellington	1.146	Extreme	1.000	0.453	0.339	0.680	0.559	0.299	C	
Roberts Avenue zone substation	Wellington	1.125	Extreme	0.658	1.000	0.730	0.964	0.182	0.451		
Marlborough_844929028	Wellington	1.108	Extreme	0.727	0.908	0.404	0.038	0.634	0.361		
Brunswick Substation	Wellington	1.099	Extreme	0.795	0.567	0.822	0.202	0.310	0.533		
Turitea zone substation	Wellington	1.099	Extreme	0.578	0.528	1.000	0.121	0.748	0.399	I	
Marlborough_840475720	Wellington	1.098	Extreme	1.000	0.751	0.913	0.798	0.407	0.541	C	
... 332 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — WELLINGTON VS FLEET			MODIFIER AVERAGES — WELLINGTON VS FLEET		
C — Continuity	0.601 vs 0.489 (+23%)		R3 Consequence	1.010	fleet 0.970 ↑
I — Infrastructure	0.664 vs 0.490 (+35%)		R4 Graph Crit.	0.997	fleet 0.978 →
S — Saturation	0.326 vs 0.472 (−31%)		R6a Restoration	0.992	fleet 0.973 →
V — Voltage	0.559 vs 0.489 (+14%)		R6b Seismic	1.220	fleet 1.097 ↑
E — Economic	0.526 vs 0.482 (+9%)		R7 Digital Read.	0.996	fleet 0.977 →
T — Transition	0.393 vs 0.454 (−14%)				

The dominant risk driver across the Wellington corridor is **T (Transition)**, averaging 0.393 against a fleet average of 0.454 — occupying 785% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.559 vs fleet 0.489 (559% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.010 (fleet: 0.970), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.220, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Wellington

342 subs · avg R = 0.790 · H:120 C:179 E:30 M:13

The region regions-level breakdown reveals significant intra-regional variation. **Wellington** leads with a mean R of 0.790 across 342 substations, while **Wellington** scores 0.790 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Wellington is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

275 Wellington substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Wellington sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

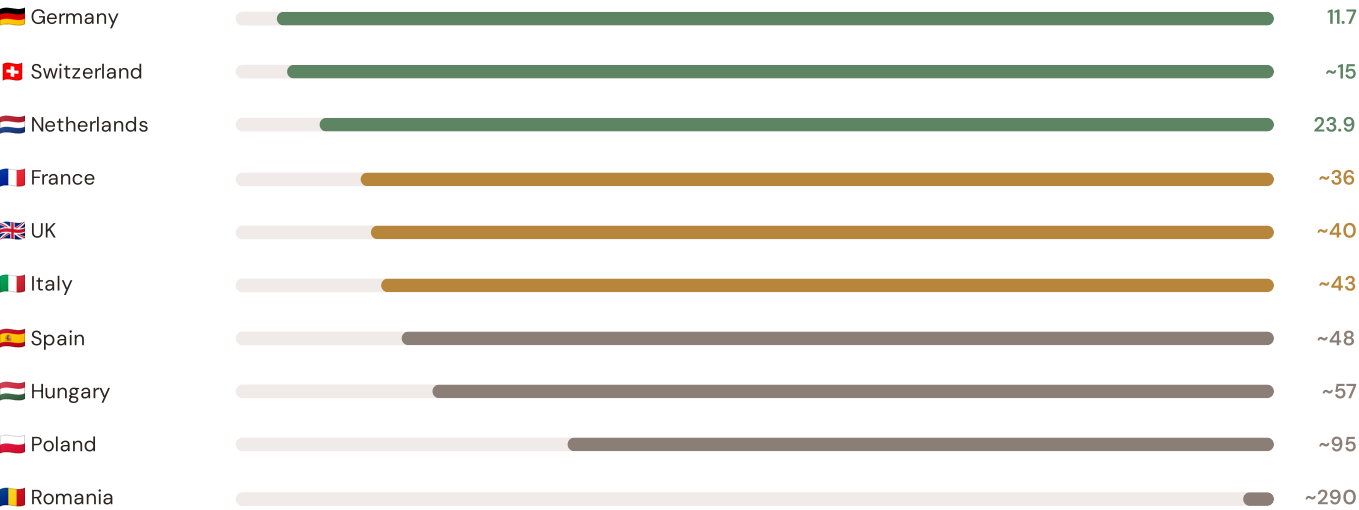
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on New Zealand's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



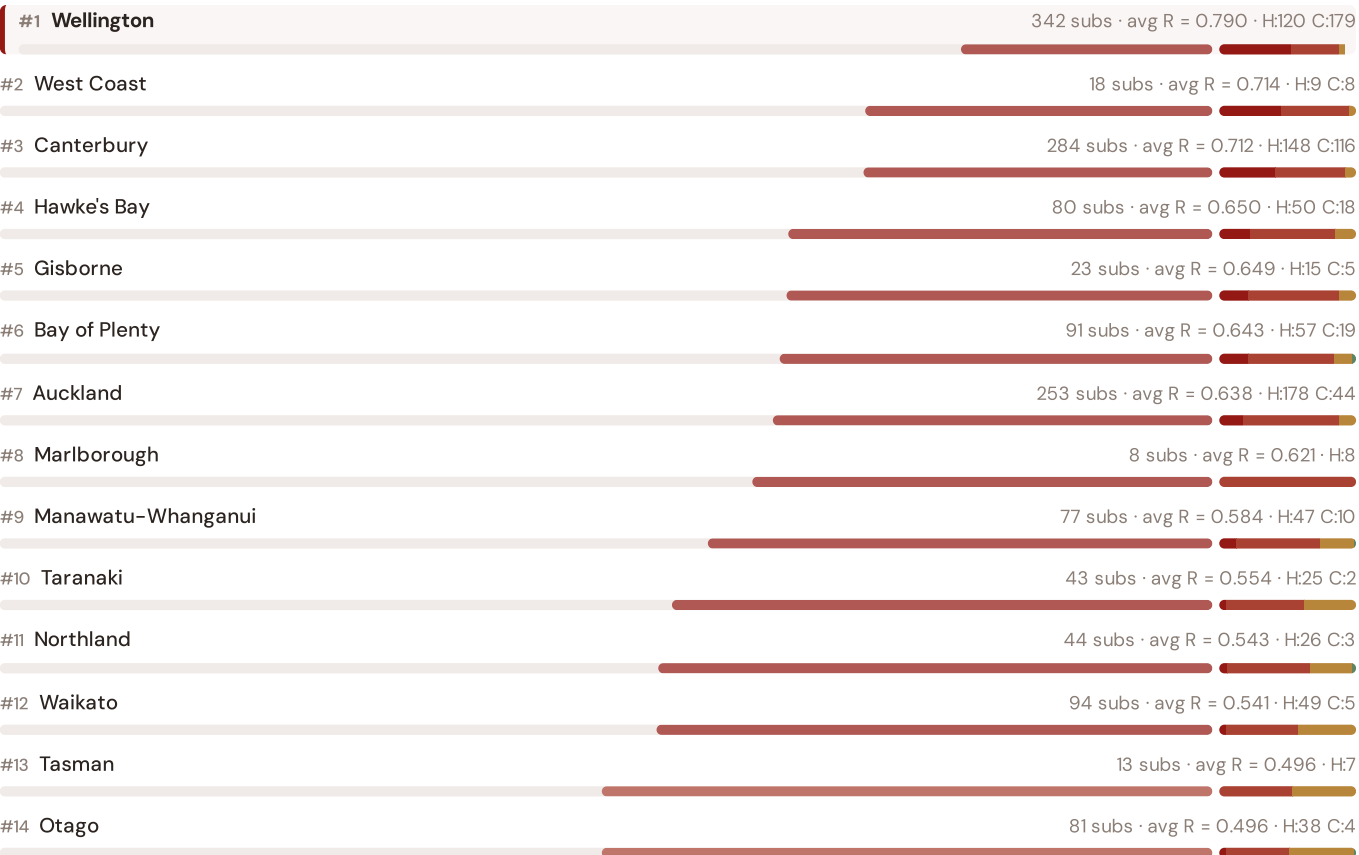
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Wellington corridor deep-dive shows **312 substations** (of 342) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.601 is **1.2× the fleet average** of 0.489. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regions Risk Ranking — All 16 New Zealand Regions

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Wellington ranks **#1 of 15 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Wellington, West Coast, Canterbury, while the three most resilient are Southland, Otago, Tasman. 6 of 15 score above the fleet average of 0.639. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

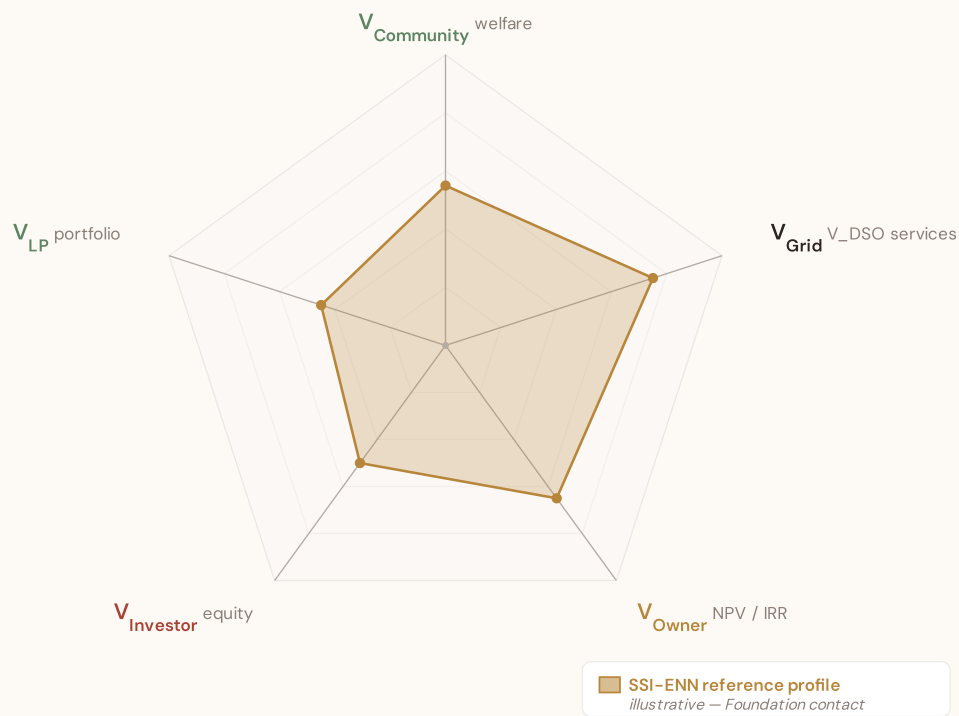
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 1 §0.3.4 · September 2026

Gaussian Copula Monte Carlo Engine

How 10,000 simulations per substation capture the correlation structure across grid risk factors

The SSI components are not independent — a substation with poor continuity (high C) often also has poor voltage quality (high V), aged infrastructure (high I5), and is located in a rural area with high economic vulnerability (high E). The Gaussian copula captures these dependencies through a 20×20 correlation matrix estimated from the national fleet, decomposed via Cholesky factorisation to generate correlated random draws. For each substation, 10,000 Monte Carlo iterations propagate uncertainty through the full scoring formula, producing not just a point estimate (R_median) but a complete posterior distribution — P5, P50, P95, confidence interval width, and skewness. This is the computational backbone, yielding the uncertainty quantification that distinguishes the SSI from deterministic indices.

KEY CONCEPTS

- 20×20 Gaussian copula with Cholesky decomposition
- 10,000 iterations per substation
- Full posterior: R_P5, R_median, R_P95, CI_width, skewness
- Correlation captures systemic clustering of risk factors

LIVE SSI DATA CONNECTION

The SSI Index currently scores **1,589 substations**. Fleet mean R = 0.639, with an average confidence interval width of 0.168 from 10,000 MC iterations per substation. The dominant component is C (Continuity) at avg 0.489 / 0.30 ceiling (163% utilisation).

Coming next month: **LAYER 5** Platform — Product Delivery Pipeline — *From raw analytical outputs to four commercial product formats*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill
W1 · W2 · W3 · W4 · W8
Peer-reviewed methodology operational at fleet scale.

Commercial (3) — dashed-faded
W6 · W7 · W10
Commercial deliverables — please contact the Foundation.

Published-baseline (1) ▲ — solid fill + chevron corner marker
W9 — Supply Security
Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill
W5 — Digital Inclusion
Foundation methodology under development; baseline value defensible at v4.2.

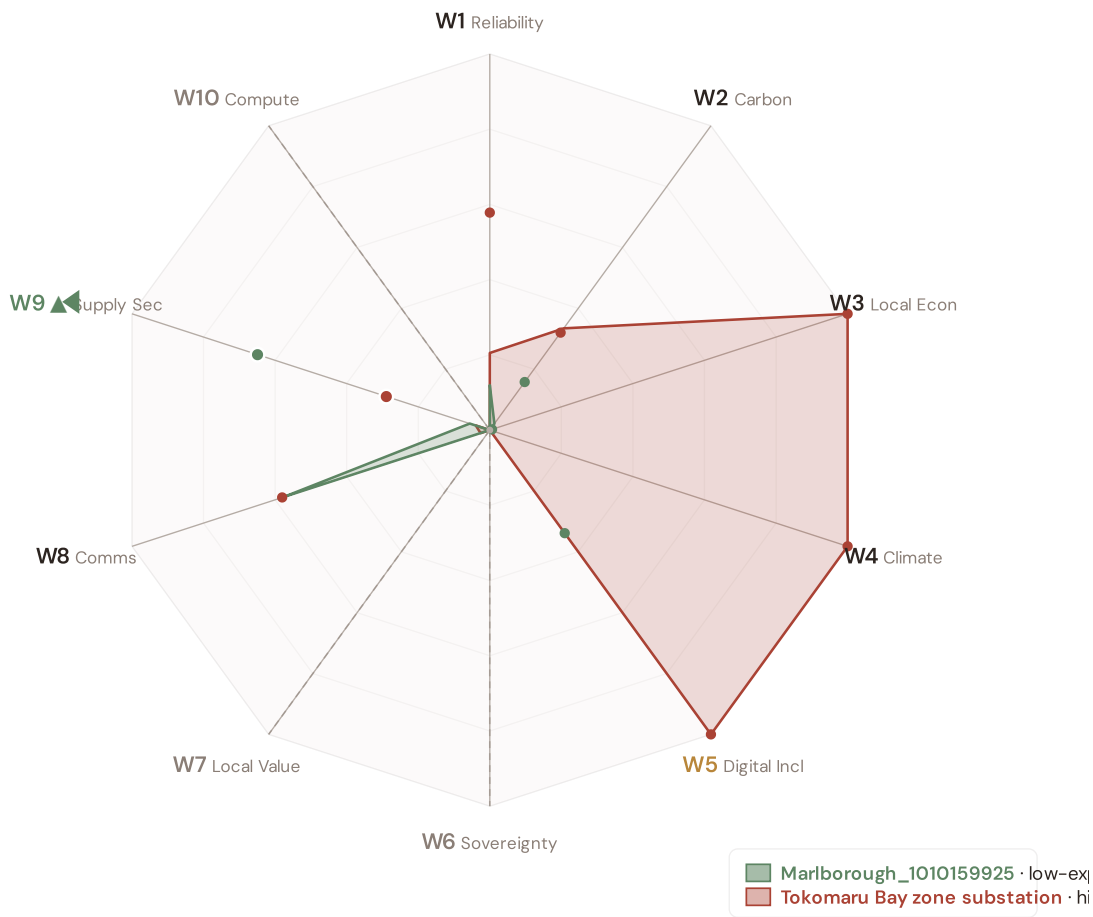
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across New Zealand, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Brunswick Substation** (low-exposure reference, Re_norm 0.000) and **[NZ_100001, 33.0 kV, West Coast]** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Marlborough_1010159925 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Tokomaru Bay zone substation polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{local} \times population (SAIDI + + -POV)$
W2	Carbon	Published	$T1 \times zonal\ MEF \times (V \cdot deg / 100) (+ EEA)$
W3	Local Economic	Published	$norm(RIO_{just}) - EU-SILC + OIPE\ energy-poverty$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2\ resilience\ modifier\ family$
W5	Digital Inclusion	Baseline-only	$norm(R8_{adapt}) - bulk\ SM + + DESI$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{crit_10yr} \times population (Markov\ 10-yr\ critical\ probability)$
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{pct} + 0.30 \times bridge + 0.10 \times (1 - deg / 20) - OSM\ power-graph\ betweenness\ centrality + bridge\ identification + degree-based\ redundancy$
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — Marlborough_1010159925 (low-exposure reference) and Tokomaru Bay zone substation (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

Looking Ahead

NEXT MONTH — EDITION 05

Alpine Valleys

Deep-dive into Region 05's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Stats NZ (Tatauranga Aotearoa)** (8 variables → Census, GDP, population, employment, household income).

FLEET SPOTLIGHT

Seismic Zone 4-5: 44 Substations in Highest Hazard

44 substations sit in the highest seismic hazard zones (Zones 4-5, moderate to strong). Of these, **29** already score High/Critical — meaning their grid condition is poor *before* considering earthquake risk. The R6b seismic modifier captures this compound vulnerability.

Edition 05 will be published on the second Thursday of October 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an New Zealand utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 006

Published 10 September 2026 · SSI Index **v4.2** (refreshed June 2026) · 1,589 substations · 76,379 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.